

Does cognitive decline make older adults misweight their health risks more?

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1 Introduction

1.1 Background

Decision-making has consistently been a prominent area of interest in the field of economics. In the economic context, decision-making typically revolves around how individuals, firms, and governments allocate limited resources under conditions of uncertainty, seeking to optimize various objectives such as utility, profit, or social welfare.

In economics, one of the most influential models of decision-making is the Expected Utility (EU) model (von Neumann and Morgenstern (1944)). The EU model posits that individuals evaluate potential outcomes by weighing the utility of each outcome against its probability, ultimately selecting the option that maximizes expected utility.

Subsequently, behavioral economics refined the EU model and introduced Cumulative Prospect Theory (CPT) (Tversky and Kahneman (1992a)). Unlike the EU model—which generally assumes consistent risk preferences and rational probability assessments—CPT incorporates empirically observed phenomena such as loss aversion and probability weighting distortions, providing a more nuanced depiction of human decision behavior.

Key factors influencing decision-making include information, individual preferences, and risk attitudes. Among these, information is particularly critical (Stigler (1961)). Access to, and comprehension of, relevant information can substantially shape decision outcomes by altering perceptions of risk and potential benefits.

Cognitive ability, which encompasses the capacity to receive, integrate, and apply information—including memory, attention, and logical reasoning—further modulates these decision processes. Individuals with higher cognitive capacity tend to assess probabilities more accurately and reason systematically, culminating in more rational choices (Stanovich and West (2000); Frederick (2005)). Conversely, those with lower cognitive ability may be more susceptible to emotional appeals, stereotypes, or extraneous “noise,” leading to decision errors.

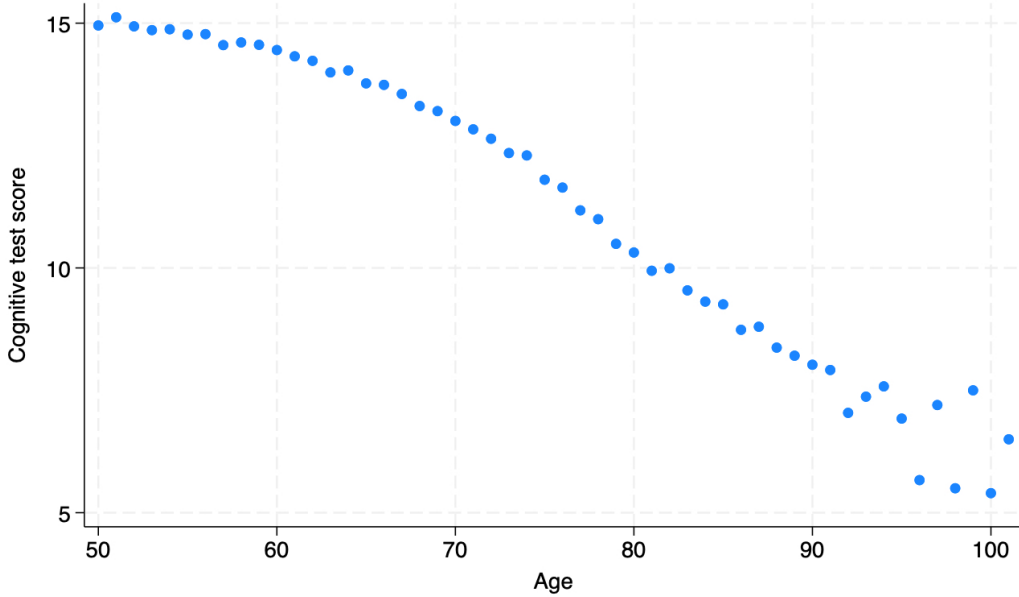
Hence, clarifying how and to what extent cognitive ability influences decision-making warrants further exploration. According to Cumulative Prospect Theory (CPT), two primary factors directly affect decisions: the *value function* and the *probability weighting function*. This study focuses on the effect of cognitive ability on the *probability weighting function*.

Because older adults exhibit a relatively rapid decline in cognitive ability (Salthouse (2004)), as illustrated in Figure 1, concentrating on this demographic is especially pertinent for the present research. Moreover, health is a crucial concern for older adults, and how they assess their personal health risks may influence a variety of decisions, including lifestyle choices, frequency of medical consultations, and the selection of health insurance. For instance, in the context of health insurance, when an older individual perceives their health risk to be higher, they may opt for more comprehen-

sive insurance plans, visit healthcare providers more frequently for preventive checkups, or adopt healthier habits in an effort to mitigate those risks.

Consequently, this study centers on how older adults’ cognitive abilities affect their assessments of personal health risks, forming the core research question of this work.

Figure 1: Relation between Cognition and Age



Source: Own calculations based on SHARE data

1.2 Relevant Literature

Enke and Graeber (2023) represents the work most closely related to the my study. Through economic experiments and mathematical analysis, the authors propose a potential explanation for the emergence of the *probability weighting function*. In particular, they argue that the function originates from *cognitive uncertainty*, which can be viewed as a measurable proxy for unobserved noisiness or heuristic-based behavior. This perspective aligns with the consensus among behavioral economists that probability weighting stems from deviations in rational processing (i.e., from noisiness or heuristics).

Like Enke and Graeber (2023), my study also investigates determinants of the *probability weighting function*. However, whereas Enke and his co-authors emphasize the concept of *cognitive uncertainty*, this study focuses on cognitive ability as a more generalized extension of that idea. As Enke and Graeber (2023) suggest, higher or lower levels of cognitive ability can substantially affect *cognitive uncertainty*. In addition, while they rely on custom experimental data, this paper makes use of the Survey of Health, Ageing and Retirement in Europe (SHARE) dataset—an existing large-scale survey.

To analyze the impact of cognitive ability on the *probability weighting function*, this study requires three core variables: (1) cognitive ability, (2) *actual* health risk probability, and (3) *subjective* health risk probability. Because this research centers on health-related decision-making, the probabilities are specified in terms of health risks. Although SHARE already provides data on cognitive ability and *subjective* health risk probabilities, it lacks direct measures of *actual* health risk probabilities. Estimation of these probabilities, therefore, is a key methodological challenge.

In this regard, Van Aken et al. (2021) offers the methodologically most similar reference. Using cutting-edge approaches based on transformer architectures—specifically, BERT—they predict clinical outcomes such as diagnoses, procedures, in-hospital mortality, and length of stay. Moreover, their data source, Medical Information Mart for Intensive Care (MIMIC) (Johnson et al. (2016)), is one of the most authoritative open datasets in the healthcare domain. By training a BERT-based model (further refined for biomedical tasks), their approach achieves strong performance in predicting health-related outcomes. However, the crucial difference from the present study lies in data type: Van Aken et al. (2021) utilizes clinical data, whereas I employ survey data from the Survey of Health, Ageing and Retirement in Europe (SHARE) (Börsch-Supan et al. (2013)).

Another challenge is that SHARE, despite containing several *cognition*-related variables, does not feature a composite measure readily interpretable as overall cognitive ability. Consequently, this paper constructs a measure of cognitive functioning by adapting the approach used in the Health and Retirement Study (HRS). The Health and Retirement Study (HRS) is a large-scale, longitudinal panel survey of individuals aged 50 and older in the United States. One of the distinguishing features of the HRS is its systematic assessment of cognitive function, which incorporates multiple tests aimed at capturing diverse cognitive domains, such as short-term memory, delayed recall, and basic arithmetic reasoning. Over the years, the HRS’s cognitive tests have undergone rigorous validation and refinement.

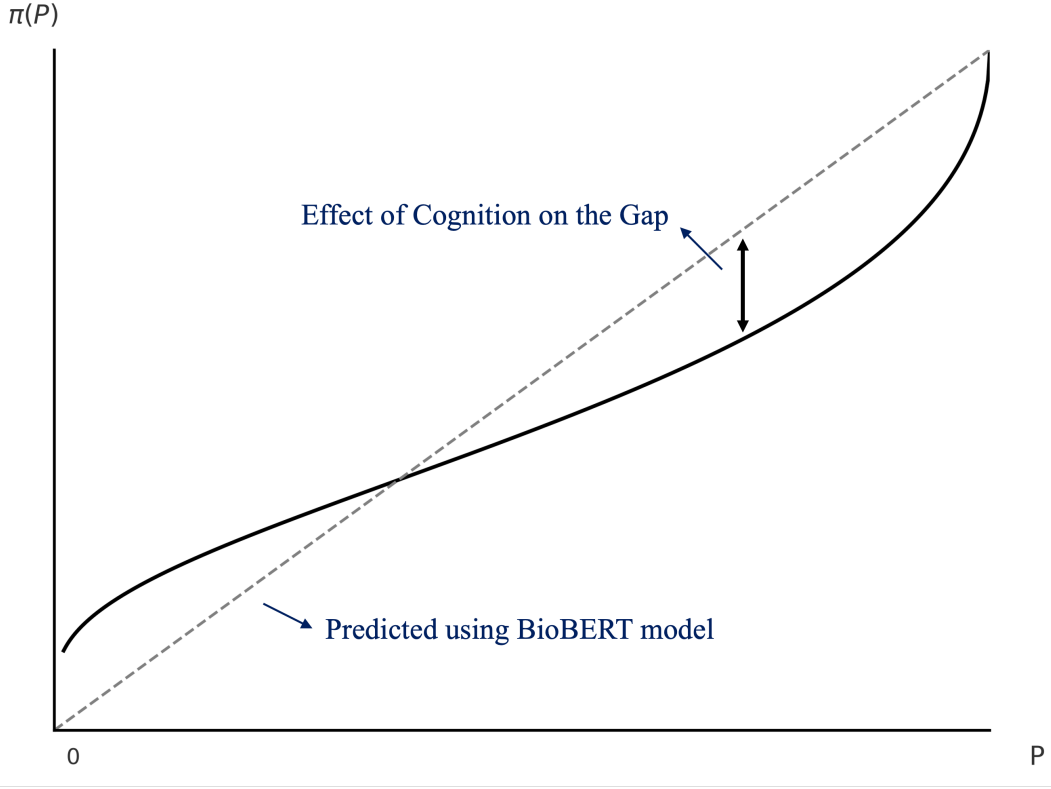
In particular, Kézdi and Willis (2014) from HRS detail four tests—two word recall tests, a counting backward test, and a serial sevens test—that together generate a composite score of 27 points. Since the majority of SHARE waves (Waves 1–7) lack data on counting backward, this study employs the remaining three tests (two word recall tests and a serial sevens test), yielding a 25-point composite measure of cognitive functioning. This adapted score thereby approximates HRS’s methodology while remaining consistent with the available SHARE data.

1.3 Framework

According to Cumulative Prospect Theory (CPT), individuals tend to *overweight* small probabilities and *underweight* large probabilities. Therefore, to address this research question, I collect both the *subjective* and *actual* health risk probabilities of older adults, and then estimate the effect of

cognition on the gap between these two probabilities (see Figure 2 for the overall framework).

Figure 2: Thesis Framework



First, I evaluate the *actual* health risks of older adults. Because the data used in this thesis (SHARE data) only include a measure of *subjective* health risks, I employ the BioBERT model to predict the *actual* health risks, obtaining relatively accurate results.

Second, using these two sets of data (*actual* and *subjective*), I plot the *probability weighting function* and provide empirical evidence supporting it.

Finally, I generate a measure of cognitive ability from the SHARE data and use a *two-way fixed effects* (TWFE) model with *clustered standard errors* to regress the gap between *subjective* and *actual* probabilities on *cognition*. The results reveal a significantly negative correlation, suggesting that higher cognitive ability is associated with a smaller gap.

The structure of this thesis is as follows: Section 2 describes the data and introduces the main variables. Section 3 presents the prediction methods and results for *actual* health risks. Section 4 provides empirical evidence for *probability weighting function*. Section 5 evaluates how *cognition* affects the gap between the two probabilities and reports the corresponding results. Section 6 concludes. Section 7 discusses possible applications of the findings.

2 Data Description

2.1 Data Source

This study utilizes data from the Survey of Health, Ageing and Retirement in Europe (SHARE) (Börsch-Supan et al. (2013)). SHARE is a multidisciplinary and cross-national longitudinal study that focuses on individuals aged 50 and older from 27 European countries (including Israel). It collects comprehensive information on health, social relationships, and economic conditions.

Currently, SHARE has 9 waves of data, collected between 2004 and 2022, with each wave occurring at approximately two-year intervals. These repeated observations enable researchers to track and analyze changes in individual health, social, and economic factors over time.

2.2 Data Summary

The analysis presented in the thesis draws on all nine waves of SHARE data, encompassing the full range of countries included in the dataset. SHARE organizes its questionnaires into multiple modules, from which this study primarily uses the following:

- **Demographics/Networks:** Provides information on age, gender, education, marital status, and social networks.
- **Physical Health:** Captures information on chronic diseases, self-reported health status, and objective health measures.
- **Health Care:** Includes data on healthcare usage, hospital stays, and out-of-pocket medical expenditures.
- **Behavioral Risk:** Focuses on lifestyle variables such as smoking, drinking, and physical activity.
- **Generated Health Variables:** Contains derived measures of health status and physical functioning, which are constructed from raw responses.
- **Coverscreen:** Collects screening information, including eligibility and response records of participants.
- **Cognitive Function:** Measures aspects of cognitive ability, such as memory, verbal fluency, and numeracy tests.

In examining the relationship between *cognition* and the gap in health risk probabilities, this study utilizes a total of 68 control variables in addition to the dependent variable (i.e., *probabilities gap*) and the key independent variable (i.e., *cognition*).

Table 1: Descriptive Statistics of the Main Variables

	Observations	Mean	SD
Probabilities Gap	114209	0.26	0.21
Cognition	114209	13.11	4.71
Body Mass Index	114057	26.47	6.08
Ever Diagnosed	114199	0.22	0.42
Taking Medication	114204	0.24	0.43
Depression Scale	113166	2.39	2.21
Exercise Frequency	114103	2.56	1.33
Limited in Activities	114203	2.42	0.72
Verbal Fluency Score	113405	20.72	7.88

Table 1 provides summary statistics for the primary variables used in this study. A total of 114,209 observations are available for most variables, except where data are missing or not collected. Variable *Probabilities Gap* represents the difference between individuals' *subjective* and *actual* health risk probabilities. A mean gap of 0.26 indicates that, on average, *subjective* probabilities exceed *actual* probabilities by roughly 26 percentage points, suggesting the presence of substantial miscalibration in risk perception among older adults. Variable *Cognition* reflects the cognitive functioning of individuals, derived from multiple tests measuring memory, verbal fluency, and other cognitive domains. The relatively high standard deviation underscores variation in cognitive abilities across the sample. The binary variables *Ever Diagnosed* and *Taking Medication* indicate whether respondents have ever been diagnosed with a major health condition, or are currently taking any prescribed medication. The mean value of 0.22 and 0.24 suggests that about three quarters of participants have had a prior diagnosis or rely on some form of ongoing pharmaceutical treatment.

3 Actual Health Risk Prediction

3.1 Empirical Design

3.1.1 Variable Transformation

To conduct a prediction of *actual* health risk, this study uses the probability of death as a proxy for health risk. The dataset employed in this research only provides a binary measure indicating whether an individual subsequently died (coded as 0 or 1), without offering a continuous measure of mortality probability. In order to construct a more precise measure of mortality probability, this study uses health-related predictors (e.g., age, chronic conditions, exercise habits, healthcare usage) and models the eventual occurrence of death as the predicted variable. By applying predictive techniques to these data, we derive individualized probabilities of death, which serve as our proxy for *actual* health risk.

3.1.2 Prediction Models

Health risk prediction has been extensively studied in the literature, and a range of methodological approaches can be employed. Broadly speaking, three major classes of models are commonly used (Mahmoudi et al. (2020)):

- **Cox Proportional Hazards Model:** This semi-parametric survival model is widely used for time-to-event data (Cox (1972)). It makes no assumptions about the baseline hazard function but assumes that covariates have a multiplicative effect on the hazard rate. The Cox model is particularly useful for estimating the effect of explanatory variables on survival time, making it a common tool for mortality prediction in epidemiological studies (Padhukasahasram et al. (2015)).
- **Bayesian Conditional Probabilities Model:** Bayesian methods incorporate prior beliefs about parameter distributions and update these beliefs in light of new evidence (i.e., observed data). When applied to mortality risk, a Bayesian framework can yield posterior probability estimates for an individual's likelihood of death, while systematically accounting for parameter uncertainty and prior information (Chen et al. (2016)).
- **Deep Learning Models:** Deep learning has gained significant traction for health risk prediction, especially in contexts involving large, complex datasets (Rajkomar et al. (2018)). Techniques such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks and, more recently, Transformer-based models (e.g., BERT) have all been effectively applied to predict mortality from electronic health records and clinical notes.

Deep Learning Architectures for Health Risk Prediction. Among the various deep learning architectures used in health-related research, CNNs (Cheung and Dahl (2018)), RNNs (Reddy and Delen (2018)), and LSTMs (Harutyunyan et al. (2019)) have been extensively studied:

- **Convolutional Neural Networks (CNN):** Originally developed for image processing tasks, CNNs can also handle structured tabular data and time-series data by leveraging convolutional filters that capture local patterns (LeCun et al. (1998)). They are particularly effective when the input features exhibit spatial or temporal structure.
- **Recurrent Neural Networks (RNN):** RNNs maintain hidden states that evolve over sequential inputs, making them suitable for modeling temporal dependencies in health data. Traditional RNNs, however, may struggle with vanishing or exploding gradients over long sequences (Schmidt (2019)).
- **Long Short-Term Memory (LSTM):** An LSTM is a special kind of RNN designed to mitigate vanishing or exploding gradients by incorporating gating mechanisms (Hochreiter and Schmidhuber (1997)). LSTMs are particularly effective for longer sequences, making them well-suited for longitudinal patient data or electronic health records.

Transformer-Based Models (e.g., BERT). In recent years, Transformer-based models have emerged as state-of-the-art in various natural language processing and predictive tasks (Vaswani et al. (2017)). The original Transformer architecture eliminates recurrence by using attention mechanisms to capture context across sequences efficiently. One of the most widely known Transformer-based models is BERT (Bidirectional Encoder Representations from Transformers), which has shown impressive performance in tasks ranging from text classification to sequence prediction (Devlin et al. (2019)).

By undergoing further specialized training on different corpora and data types, BERT has been adapted into a variety of model architectures, such as ELECTRA (Clark et al. (2020)), XLNet (Yang et al. (2019)), DistilBERT (Sanh et al. (2019)), SPANBERT (Joshi et al. (2020)), and BERTSUM (Liu and Lapata (2019)). These variants introduce modifications designed for efficiency, improved language modeling, or enhanced performance on specific tasks.

BioBERT. BioBERT is a domain-specific variant of BERT trained on large-scale biomedical corpora, such as biomedical articles and clinical records (Lee et al. (2020)). By incorporating domain-specific vocabulary and syntax, BioBERT has been shown to achieve superior performance in a variety of biomedical and clinical tasks, including named entity recognition, relation extraction, and health risk prediction.

Given its strong empirical performance in predicting health-related outcomes, BioBERT is selected for this study to estimate mortality risk. This choice is motivated by its demonstrated ability

to capture nuanced relationships among clinical and demographic variables, ultimately yielding more accurate predictions of individual health risk profiles.

3.1.3 Lack of Clinical Data

The SHARE dataset is based on survey interviews, collecting self-reported information from older adults. Although it contains a broad range of variables across diverse domains, it lacks clinical data (such as detailed blood test results, MRI findings, and specific medication names and dosages). To mitigate this limitation and enhance the accuracy of the mortality probability predictions, this study first performs BioBERT pre-training using the MIMIC database.

MIMIC (Medical Information Mart for Intensive Care) is a large, publicly available database comprising de-identified health-related data (Johnson et al. (2016)). It contains detailed clinical information of patients who stayed in critical care units, including demographics, vital signs, laboratory tests, medications, and procedures. By pre-training BioBERT on MIMIC, the model learns domain-specific language patterns, thereby improving its capacity to capture nuances in clinical and biomedical data.

Following the pre-training phase on MIMIC, this research then trains BioBERT on the SHARE data using variables that are similar to those found in MIMIC. These variables include: demographics/networks (e.g., marital status, country of birth, and educational attainment), physical health (e.g., longstanding illness or disability, eyesight and hearing conditions, specific diagnoses, symptoms, pain, and difficulties with activities of daily living), health care (e.g., recent doctor visits, hospital stays, and health insurance status), behavioral risk (e.g., smoking status, alcohol consumption, and frequency of physical exercise), generated health variables (e.g., mental health indicators, grip strength, and walking speed).

3.1.4 Unbalanced Data

One challenge in the SHARE dataset is the relatively low observed mortality rate. For instance, among individuals who were alive at the time of a given interview, approximately 5% died within the subsequent four-year period, which leads to an unbalanced class distribution (i.e., many more survivors than deaths).

An unbalanced dataset can bias the learning algorithm toward predicting the majority class (survival) more frequently. This may compromise the sensitivity of the model in identifying those at higher risk of mortality, resulting in elevated false negatives and, in turn, diminished clinical utility (Wallace et al. (2011)).

To address this imbalance, the class weight technique can be used (King and Zeng (2001)). In this approach, the loss function in model training is weighted inversely to class frequencies,

effectively penalizing misclassifications of the minority class (death) more than misclassifications of the majority class (survival). By doing so, the model is encouraged to pay closer attention to minority-class examples.

Additionally, the MIMIC dataset also has an unbalanced mortality rate of about 5%. Therefore, in this study, a similar four-year mortality horizon is adopted for SHARE, resulting in an equivalent mortality rate of roughly 5%. Specifically, for each wave in SHARE, the mortality outcome from two waves later is merged with predictors from the earlier wave. For instance, individuals in Wave 1 have their baseline predictors matched to the *deceased* variable in Wave 3. Consequently, only the predictors from Waves 1 to 7, along with mortality indicators from Waves 3 to 9, are utilized in the analysis.

3.1.5 Data Split

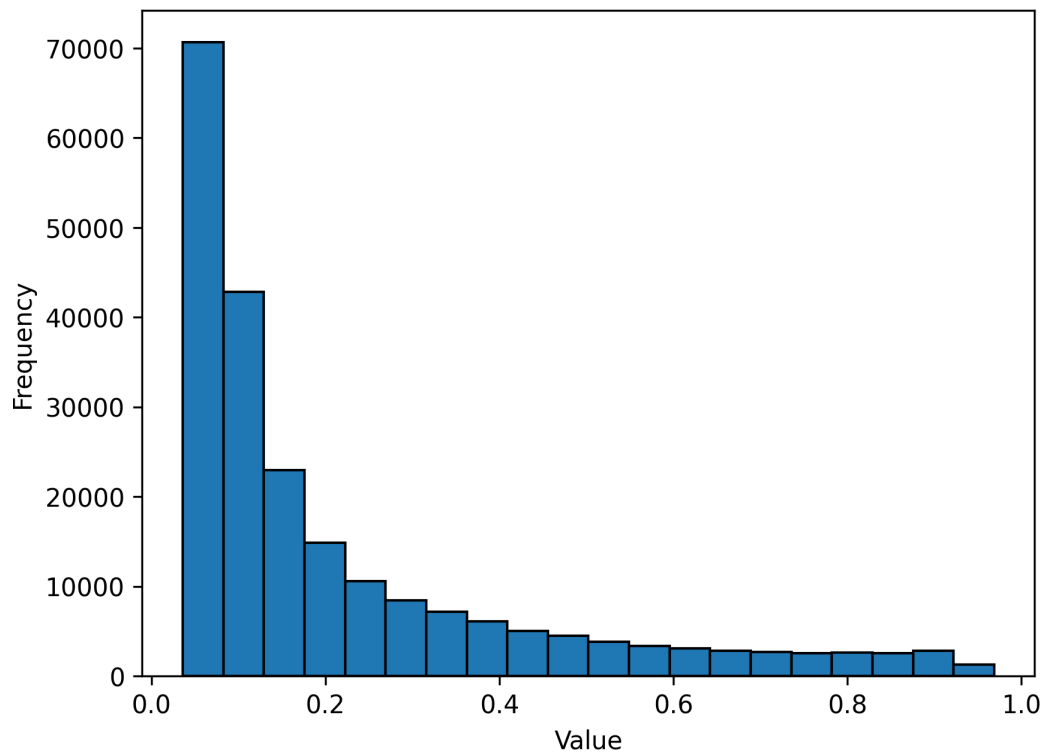
To construct training inputs that mirror the format of the MIMIC data, this study retains the labels and value labels of all predictor variables (as well as the names of their respective modules). A single text variable is then generated for each participant, organized into five major domains (demographics/networks, physical health, health care, behavioral risk, and generated health variables). Within each domain, the variable labels and their corresponding values are listed line by line. The *deceased* variable (taking values 0 or 1) is subsequently merged to these text variables according to each respondent’s wave and ID.

BioBERT requires three datasets for training: training, validation, and testing. In this study, the sample is randomly partitioned into training (70%), validation (10%), and test (20%) sets. This split enables both hyperparameter tuning and the evaluation of out-of-sample performance, ensuring that the final model generalizes effectively to unseen data.

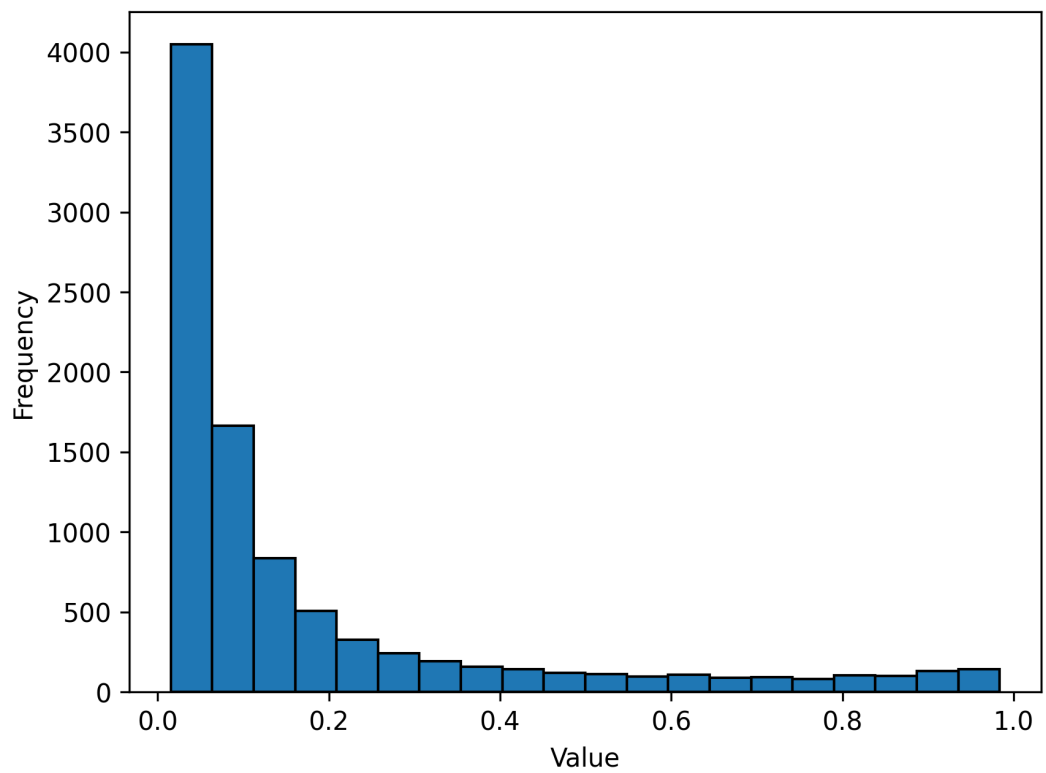
3.2 Results

Figure 3 illustrates the predicted probability distributions of mortality derived from BioBERT for both the SHARE and MIMIC datasets. In both panels, the histograms are heavily skewed toward lower predicted probability values, reflecting the relatively low actual prevalence of mortality events in each sample (i.e., class imbalance). However, the SHARE data (Figure 3(a)) exhibits a larger number of extremely low predictions (i.e., near zero) compared to the MIMIC data (Figure 3(b)). This difference likely stems from the fact that SHARE relies largely on self-reported and non-clinical variables, resulting in greater uncertainty around mortality risk for many participants. Conversely, MIMIC contains richer clinical information, enhancing the model’s ability to discriminate higher-risk cases.

Figure 3: Predicted Probability Distribution of Mortality



(a) SHARE Data



(b) MIMIC Data

3.3 Evaluation

Table 2 summarizes the performance metrics for BioBERT on the SHARE and MIMIC datasets. Despite both models attaining relatively high overall accuracy (86.83% for SHARE; 88.66% for MIMIC), closer inspection reveals disparities in class-specific performance.

Table 2: Summary of Evaluation Metrics

Metric	SHARE (%)	MIMIC (%)
Loss	67.46	59.86
Accuracy	86.83	88.66
ROC-AUC	76.90	84.07
ROC-AUC (Weighted)	76.90	84.07
F1 Macro	59.54	62.64
F1 Micro	86.83	88.66
F1 (Class 0)	92.77	93.81
F1 (Class 1)	26.30	31.46
MCC	22.80	28.57

The accuracy metric shows that the model correctly classifies more than 85% of instances in both datasets, while the micro F1-score (which weights classes by support) aligns closely with accuracy. This result is not surprising, given the low mortality rate and the model’s strong performance on the majority (surviving) class.

For SHARE, the F1-score for Class 0 (survivors) is high (92.77), whereas for Class 1 (deceased) it is only 26.30. Although the MIMIC results are similarly imbalanced, the F1-score for Class 1 is slightly higher (31.46), suggesting that richer clinical data contributes to improved sensitivity in detecting individuals at risk of mortality. This discrepancy indicates that while the SHARE-based model excels at identifying survivors, it struggles to detect less frequent mortality events with the same level of accuracy.

The Area Under the Curve (AUC) of Receiver Operating Characteristic (ROC) for SHARE (76.90) trails behind MIMIC (84.07), but the difference between the two datasets is not significant, signifying a good ability to rank-order individuals by mortality risk.

The skew in class distribution is evident in both datasets. However, since SHARE relies less on clinical measurements, the feature set may not capture sufficient prognostic details to correctly identify higher-probability mortality cases. This likely contributes to the lower F1 for Class 1 and the overall lower MCC compared to MIMIC, where additional medical variables help the model discriminate risk more effectively.

Tables 3 and 4 present the full classification reports, detailing precision, recall, F1-scores, and support for each class. For SHARE, Class 1 (deceased) has a recall of 44.84%, indicating that although the model identifies less than half of actual mortality cases, it performs significantly better in recall than in precision (18.61%). This gap points to a relatively high false-positive rate for Class 1. In contrast, MIMIC achieves a recall of 50.00% for mortality, coupled with a slightly higher precision (22.96%), underscoring the impact of clinically detailed features.

Table 3: Classification Report - SHARE Data

Class	Precision	Recall	F1-Score	Support
0	96.69	89.15	92.77	209354
1	18.61	44.84	26.30	11580
Accuracy			86.83	220934
Macro Average	57.65	66.99	59.54	220934
Weighted Average	92.60	86.83	89.29	220934

Table 4: Classification Report - MIMIC Data

Class	Precision	Recall	F1-Score	Support
0	97.06	90.78	93.82	8807
1	22.96	50.00	31.47	484
Accuracy			88.66	9291
Macro Average	60.01	70.39	62.64	9291
Weighted Average	93.20	88.66	90.57	9291

Overall, these findings imply that although the SHARE-based model is constrained by class imbalance and a lack of high-resolution clinical measures, it remains robust for identifying survivors, as on the MIMIC dataset. Therefore, the predicted probabilities are reliable and can be use as the *actual* health risk probabilities.

4 Probability Weighting Function for Health Risks

4.1 Theoretical Foundation

4.1.1 Decision Making Model in Economics

Decision-making under risk has long been a focal point in economics, particularly in the context of consumer choice and insurance markets. The Expected Utility (EU) framework, rooted in the work of von Neumann and Morgenstern (1944), posits that individuals evaluate risky prospects by considering the expected value of the utility of outcomes rather than the outcomes themselves. Formally, if an individual faces a set of possible outcomes x_i (each with probability p_i), the expected utility is given by:

$$EU = \sum_{i=1}^n p_i u(x_i),$$

where $u(\cdot)$ is a utility function capturing the individual's risk preferences.

In health economics, the EU model provides a foundation for understanding how people make health-related choices—such as whether to purchase health insurance, undergo preventive care, or follow treatment regimens. A key assumption of the EU model is that decision-makers weigh probabilities in a linear and objective manner, meaning they accurately translate the *actual* probability of outcomes (e.g., contracting an illness) into their utility calculations.

4.1.2 Cumulative Prospect Theory (CPT)

While the EU model has been central to economic analysis for decades, a growing body of evidence has revealed systematic deviations from its predictions. Cumulative Prospect Theory (CPT), introduced by Tversky and Kahneman (1992b) as an extension of Kahneman and Tversky (1979)'s Original Prospect Theory, addresses many of these discrepancies by incorporating behavioral elements such as loss aversion and reference dependence.

CPT retains the notion that individuals evaluate outcomes relative to a reference point (often the status quo or current wealth) and exhibit different risk attitudes for gains versus losses. However, the critical departure from the EU framework lies in *probability weighting*: CPT allows for the possibility that individuals distort *actual* probabilities—often overweighting small probabilities and underweighting moderate or large probabilities. In the domain of health risks, such distortions can manifest in the tendency to either overestimate the likelihood of rare illnesses (leading to over-purchasing of certain insurance products) or underestimate common risks (potentially delaying preventive measures).

Formally, a prospect P can be expressed as a set of possible outcomes $\{x_1, x_2, \dots, x_k\}$ with associated probabilities $\{p_1, p_2, \dots, p_k\}$. CPT typically partitions these outcomes into gains ($x_i > 0$)

and losses ($x_j < 0$), with a *value function* $v(\cdot)$ that is concave for gains and convex for losses, and often steeper in the loss domain to capture loss aversion. The *probability weighting functions* $w^+(p)$ and $w^-(p)$ then transform *actual* probabilities into *subjective* decision weights for gains and losses, respectively. A simplified version of the CPT “value” of prospect P can be written as:

$$V(P) = \sum_{i \in \text{gains}} w^+(p_i) v(x_i) + \sum_{j \in \text{losses}} w^-(p_j) v(x_j).$$

Here,

- $v(\cdot)$ is the *value function*, usually S-shaped around a reference point (e.g., zero change from the status quo) and asymmetric to reflect loss aversion.
- $w^+(\cdot)$ and $w^-(\cdot)$ are the *probability weighting functions* for gains and losses, respectively, often exhibiting an inverse-S shape.

4.1.3 Probability Weighting Function

A key component of CPT is the *probability weighting function*, which transforms *actual* probabilities into decision weights. Rather than treat probabilities linearly, individuals employ a weighting function $w(\cdot)$, such that an *actual* probability p is mapped into a *subjective* probability $w(p)$. Common functional forms for $w(\cdot)$ proposed in the literature include:

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}},$$

where the parameter γ (or sometimes two parameters) governs the curvature of the weighting function. Experimental and empirical research suggests that $w(p)$ typically exhibits *inverse-S* shape: for small probabilities, $w(p) > p$, indicating overweighting; for larger probabilities, $w(p) < p$, indicating underweighting.

Typically, the *probability weighting function* is attributed to either “noisiness” or a “heuristic” dimension. Heuristics refer to mental shortcuts or rules of thumb that simplify decision-making, often at the cost of systematic biases or departures from fully rational behavior. “Noise,” in this context, can be understood as random variation in how individuals perceive or process probabilities, leading to inconsistencies in their judgments.

Enke and Graeber (2023) propose a measurable proxy for this noisiness or heuristic tendency, known as “*cognitive uncertainty*”. They define *cognitive uncertainty* as the subjective uncertainty a person holds regarding the optimal choice of action in their own decision-making. Concretely, if an individual is more uncertain about how to interpret probabilities (or conceptualize risk) systematically, she tends to “blur” or “compress” these probabilities toward 50% on a systematic

basis. In more intuitive terms, if someone is unsure about the true likelihood of an event—even if informed that it is 80%—she may mentally represent it as something closer to 65% or 70%. Conversely, if she is told it is only 20%, she might “round” it upward to 30% or 35%. This phenomenon narrows the *subjective* gap between high and low probabilities.

In the realm of health economics, the *probability weighting function* has been increasingly applied to understand individuals’ *subjective* assessments of morbidity and mortality risks. For instance, older adults facing uncertainty about chronic diseases or life expectancy may systematically *overweight* small risks of catastrophic outcomes (e.g., a rare but severe disease) and *underweight* more common yet less dramatic risks. Such distortions can result in suboptimal behaviors, such as the underuse of preventive care or the overuse of expensive diagnostic tests for rare conditions.

Overall, the introduction of probability weighting into decision-making models provides a richer framework for analyzing how real-world individuals perceive and respond to health risks. This approach highlights the potential need for policies and interventions—such as better risk communication and targeted health education—that address misperceptions and help align *subjective* risk assessments with *actual* probabilities.

4.2 Results

4.2.1 Subjective Health Risk

As described in Section 3, the *actual* health risk probabilities are predicted using the BioBERT model, drawing on health-related variables and the corresponding mortality status four years later. By contrast, the *subjective* health risk probabilities are derived from participants’ responses to a specific question in the Expectation module:

“What are the chances that you will live to be age [80/85/90/95/100/105/110/120] or more? I’d like for you to give me a number from 0 to 100.”

The precise target age in this question varies according to each respondent’s current age, as detailed in Table 5. Roughly speaking, for respondents currently under 65, the target age is 75 (i.e., approximately ten years into the future); for those over 80, a higher target age is used (e.g., 95 or 100), and so forth.

Table 5: Mapping of Age to Target Age

Age	Target Age
< 65	75
65 - 69	80
70 - 74	85
75 - 79	90
80 - 84	95
85 - 94	100
95 - 99	105
100 - 104	110
> 104	120

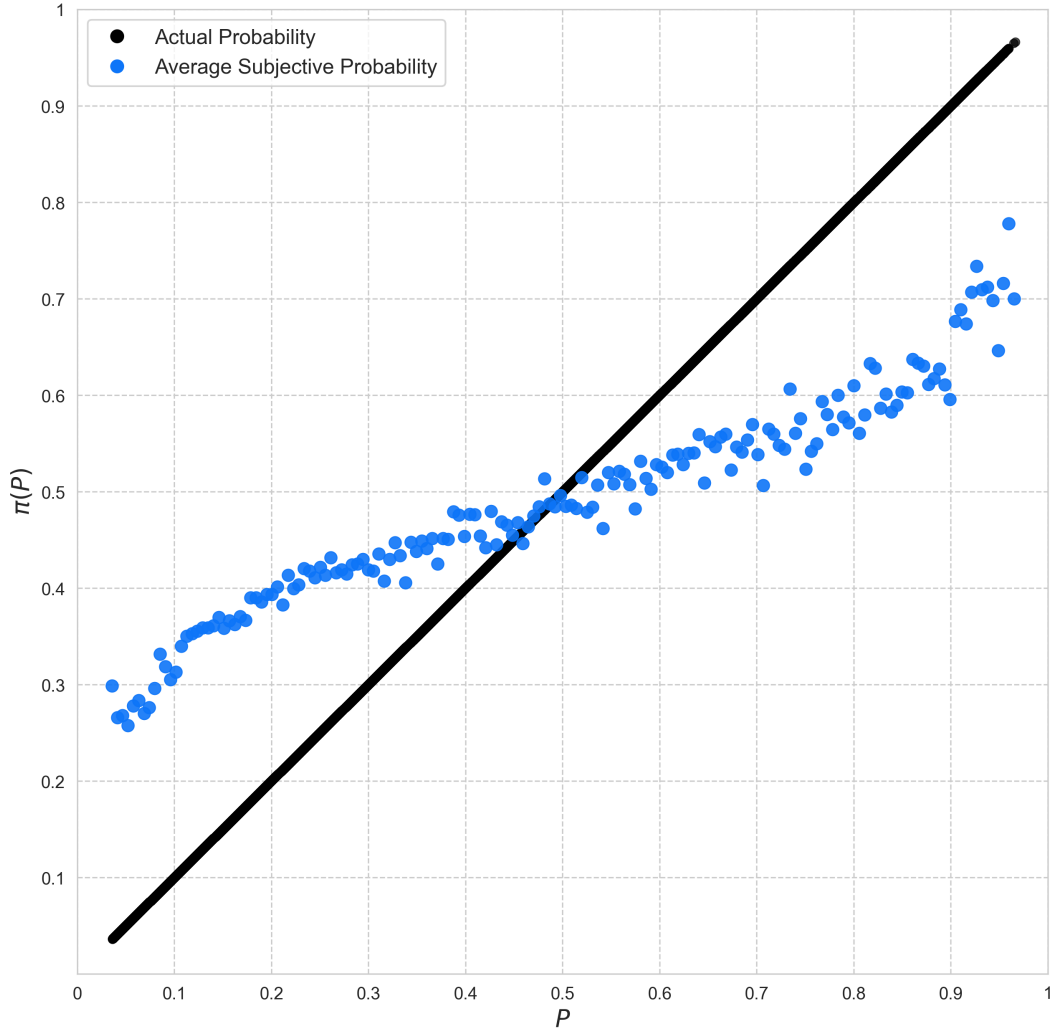
In practice, each respondent is asked for the *subjective* probability (from 0% to 100%) that they will be alive at the respective target age. To convert this into the *subjective* probability of dying by that age, the present study calculates $1 - (\text{response}/100)$. Hence, if a participant reports a 70% chance of being alive at age 80, their *subjective* mortality risk by that age is $1 - 0.70 = 0.30$.

4.2.2 Empirical Results

Figure 4 compares two probabilities for older adults:

- $\pi(P)$ (blue points): the average *subjective* health risk probability reported by individuals (Because there are more than 220,000 data, Figure 4 presents only representative average *subjective* health risk probabilities), and
- P (black line/dots): the *actual* health risk probability based on the BioBERT model.

Figure 4: Subjective and Actual Health Risk Probabilities



In the lower range of P (approximately 0 to 0.3), the blue dots generally lie above the diagonal $y = x$. This suggests that for relatively small risks, individuals tend to overestimate their likelihood of mortality. Such behavior is consistent with the concept of “*overweighting*” small probabilities, as formalized in Cumulative Prospect Theory (CPT). Practically, older adults might be disproportionately worried about comparatively unlikely health events, leading to an inflated perception of risk.

Around the middle range of P (roughly 0.4 to 0.6), the *subjective* and *actual* probabilities converge more closely, with blue points clustering near the diagonal. This indicates that, on average, participants’ *subjective* assessments are more accurate when the underlying risk is neither too low nor too high. The tighter clustering here implies that moderate likelihoods are perceived with relatively less distortion, although small deviations remain evident.

For P above about 0.7, the blue dots often lie below the diagonal, implying that older adults

underestimate very high probabilities of mortality. According to probability weighting theory, people commonly *underweight* large risks—perhaps due to psychological defense mechanisms, denial, or other cognitive biases that lead them to minimize the perceived severity of near-certain events.

Taken together, the overweighting of small probabilities and underweighting of large probabilities generate a pattern resembling an *inverse-S* curve. This shape is a hallmark of CPT’s *probability weighting function*, in which *subjective* probability deviates systematically from *actual* estimates. Such findings align with extensive literature in behavioral economics showing that individuals tend to distort probabilities, with crucial consequences for health-related behaviors (e.g., purchasing health insurance, undergoing routine screenings).

Although the points form a recognizable trend, there is noticeable dispersion around the main curve (especially in the moderate range). This dispersion suggests heterogeneity in how individual respondents assess their personal risk. Some older adults may be overly pessimistic, while others may be markedly optimistic, even if their *actual* risk profiles are similar.

4.2.3 Different Scales for the Two Probabilities

One crucial caveat is that the *actual* health risk measure captures mortality status after four years, whereas the *subjective* measure references roughly a 10 to 15 year horizon. Due to data limitations, it was not feasible to use a 10 to 15 year span for the *actual* risk, as this would drastically reduce the training sample for the BioBERT model and thus compromise predictive accuracy.

Although the time horizons for the two probabilities differ, it is reasonable to assume that the shape of the weighting function would remain similar if both probabilities referred to the same time window. Specifically, if the *actual* risk probabilities were estimated for a 10 to 15 year period, the overall curve might shift rightward (since mortality risk would naturally increase over a longer horizon). However, the relative pattern of underweighting and overweighting probabilities is not expected to change drastically.

From a theoretical perspective, individuals’ *subjective* assessments often deviate from the *actual* ones due to cognitive biases, incomplete information, or personal experience. Consequently, the general patterns visible in Figure 4 are broadly consistent with the *probability weighting function* in behavioral economics—most notably, Cumulative Prospect Theory—where small probabilities are often exaggerated while higher probabilities are downplayed.

Despite the discrepancy in the length of the forecast windows, this figure illustrates that older adults may hold systematically different perceptions of their own health risks compared to model-based predictions. These gaps have implications for policy interventions aimed at better aligning *subjective* beliefs with *actual* risk, potentially improving health planning and retirement decision-making among older populations.

5 Effect of Cognition on the Gap of Health Risks

5.1 Empirical Design

5.1.1 Dependent Variable

To investigate the effect of cognitive ability on older adults' biases in evaluating their own health risks, this study takes the gap between *subjective* and *actual* health risk probabilities as the dependent variable. As shown in Figure 4, older adults sometimes overestimate certain risk probabilities and underestimate others. However, this research is only concerned with the magnitude of the *subjective* misestimation, not whether it is positive or negative. Therefore, the absolute difference between the two probabilities $|\pi(P) - P|$ is used as the dependent variable.

5.1.2 Variable of Interest

Cognitive function, often viewed as a critical determinant of decision quality, serves as the key explanatory variable. In the literature, cognitive ability is frequently divided into two broad categories: fluid intelligence and crystallized intelligence (Horn and Cattell (1966)). Fluid intelligence refers to the capacity for logical reasoning, pattern recognition, and problem-solving in novel situations. It underpins our ability to manipulate and process information rapidly. Crystallized intelligence, by contrast, comprises accumulated knowledge and skills over one's lifetime—such as vocabulary, factual information, and culturally embedded expertise—and generally remains more stable as individuals age.

Empirical studies often break down cognitive function into multiple domains (e.g., memory, executive function, verbal fluency, processing speed), each measured by specialized tasks (Diamond (2013)). For instance, working memory can be tested via digit span or word recall tasks, while executive function is commonly measured by tests of task-switching, response inhibition, or problem-solving. Processing speed—the rate at which an individual can perceive, understand, and respond to information—often decreases with age and is considered a key driver of the overall decline in fluid intelligence.

These distinctions in cognitive function have practical implications for older adults' decision-making across domains such as finance, healthcare, and social interaction. A decline in fluid intelligence might, for example, lead to difficulties in comparing complex health insurance plans or adhering to treatment regimens. In contrast, crystallized intelligence may help older adults compensate by relying on a lifetime of accumulated knowledge and experience to make choices. Understanding these patterns is therefore crucial for researchers and policymakers seeking to identify potential interventions—such as cognitive training or health education programs—that could mitigate the effects of cognitive decline on important life decisions.

In this study, cognitive functioning was primarily used to measure fluid intelligence. Specifically, it was measured by a 25-point composite variable that combined performance on three tests:

- **Immediate Word Recall.** Participants are presented with a list of 10 words (e.g., Goat, Door, Fish, Desk, Rope, . . .) and asked to recall as many as they can immediately. Each correctly recalled word is awarded one point, so the maximum score for this test is 10 points.
- **Delayed Word Recall.** After a short interval (during which participants perform another task to prevent rehearsal), they are asked to recall the same 10 words again (delayed recall). The scoring logic mirrors that of the immediate recall test: each correct answer is worth one point, up to a maximum of 10 points.
- **Serial Sevens Test.** Participants are instructed to subtract 7 repeatedly from a given number (e.g., from 100) and to state each subsequent result in sequence. They perform five subtractions in total; each correct subtraction yields one point, giving a maximum of 5 points for this task.

Summing these three components (10 points from immediate recall, 10 points from delayed recall, and 5 points from serial sevens) yields a maximum composite score of 25. The emphasis on these particular tasks reflects their ability to capture distinct cognitive processes—short-term memory, delayed recall (which involves encoding and retention), and basic arithmetic reasoning under time or memory constraints.

Under typical circumstances, this cognitive scale has a maximum of 27 points because it also includes a counting backward test (Kézdi and Willis (2014)). The counting backward test requires participants to recite a sequence of numbers in reverse order (e.g., from 20 down to 1). However, since most SHARE waves lack data for this test (only Wave 8 and Wave 9 contain this variable), it is excluded to maintain consistency across all observations, reducing the total composite score to a 25-point scale for the present analysis.

5.1.3 Model Selection

Researchers have a variety of econometric methods at their disposal for causal inference (Hansen (2022)), including:

- **Ordinary Least Squares (OLS) with Control Variables.** OLS remains one of the most commonly employed methods for estimating linear relationships. In a causal inference setting, researchers include a rich set of control variables to account for observable confounders. Although simple to implement, OLS cannot fully address endogeneity issues arising from unobservable factors or reverse causality without additional assumptions or techniques (e.g., fixed effects, panel data, etc.).

- **Instrumental Variables (IV) Estimation.** IV methods help address endogeneity by introducing an external source of variation—an “instrument”—that affects the dependent variable only through its impact on the endogenous regressor (Angrist and Krueger (2001)). If the instrument is both relevant (correlated with the regressor) and valid (exogenous to the error term), the IV estimator can yield consistent causal estimates even when omitted variable bias or measurement error exists.
- **Differences-in-Differences (DiD).** DiD exploits variation across groups and time to estimate causal effects. Typically, one group undergoes a policy intervention (treatment) while a comparable group does not (control). By measuring outcomes before and after the intervention, DiD subtracts out common trends, isolating the policy’s causal impact under the key assumption that both groups would have followed parallel trends in the absence of treatment (Bertrand et al. (2004)).
- **Propensity Score Matching (PSM).** PSM reduces selection bias by matching treated units to control units with similar observed characteristics. After matching, the outcomes in the treated group can be compared to those in a closely resembling control group, approximating a randomized experiment. However, PSM relies on the “selection on observables” assumption—meaning any relevant confounders must be observed and included in the matching algorithm (Rosenbaum and Rubin (1983)).
- **Fixed Effects Models.** Fixed effects models exploit repeated observations (panel data) of the same units (e.g., individuals, firms) over time to control for unobserved, time-invariant characteristics. By differencing or demeaning the data, these models remove any confounding factors that do not vary over time. This allows for more robust causal inference, provided that key identifying assumptions (e.g., changes in the outcome over time are not driven by omitted time-varying confounders) are satisfied.

In this study, a *two-way fixed effects* (TWFE) model with *clustered standard errors* is chosen to estimate the causal relationship between the health risk gap and cognitive function. The *two-way fixed effects model* includes both *individual and time fixed effects*, controlling for time-invariant unobserved heterogeneity across individuals (e.g., innate traits, early-life environment) and for common shocks across time periods (e.g., macroeconomic or policy changes). This design is particularly suitable for longitudinal data like SHARE, where repeated observations on the same individuals over multiple waves allow the researcher to difference out unobserved, time-invariant confounding factors.

Formally, the *two-way fixed effects* regression can be specified as:

$$\text{Gap}_{it} = \beta \text{Cognition}_{it} + \mathbf{X}_{it} \gamma + \alpha_i + \delta_t + \varepsilon_{it},$$

where Gap_{it} is the absolute difference between *subjective* and *actual* health risk probabilities for individual i at time t ; Cognition_{it} is the cognitive score; $\mathbf{X}_{it}$ is a vector of control variables (e.g., age, gender, education, health status); α_i captures individual fixed effects; δ_t captures time (wave) fixed effects; and ε_{it} is the error term. A significantly negative β would imply that higher cognitive function is associated with smaller health risk misestimation, whereas a significantly positive β would suggest that higher *cognition* increases the gap.

By controlling for unobservable individual characteristics that do not change over time, as well as for wave-specific shocks, the *two-way fixed effects* model mitigates many potential sources of endogeneity. Thus, it provides a more rigorous basis for concluding that changes in cognitive ability lead to changes in the magnitude of older adults' *subjective* misestimation of health risks.

Besides, I employ the Hausman test (Hausman (1978)) to compare the coefficients from a fixed effects (FE) model and a random effects (RE) model. As shown in the output, the large chi-square statistic ($\chi^2 = 272.27, p < 0.001$) rejects the null hypothesis that FE and RE estimates do not differ systematically. This result indicates that the random effects model is inconsistent, while the fixed effects estimates are consistent under the maintained assumptions. Thus, the FE approach is more suitable for identifying causal effects of *cognition* on the *probabilities gap*.

In addition, robust standard errors are used to address heteroskedasticity and first-order autocorrelation. The results and analysis of assumption testing can be seen in Section 5.2.

5.2 Assumptions Testing

In this section, I examine the core assumptions underlying the *two-way fixed effects* model and discuss how test results inform the final estimation strategy.

5.2.1 Linearity

A linear functional form is central to most fixed effects regressions. The linearity assumption posits that the outcome variable (i.e., the absolute gap in health risk probabilities) is a linear function of the explanatory variables (e.g., *cognition*) plus an error term. This implies that the partial effect of *cognition* on the gap is constant across values of *cognition*.

The linearity assumption facilitates straightforward interpretation of coefficients as constant marginal effects. Violations could lead to biased or inconsistent estimates if crucial nonlinear terms are omitted.

Figure 5: Relation between Probabilities Gap and Cognition

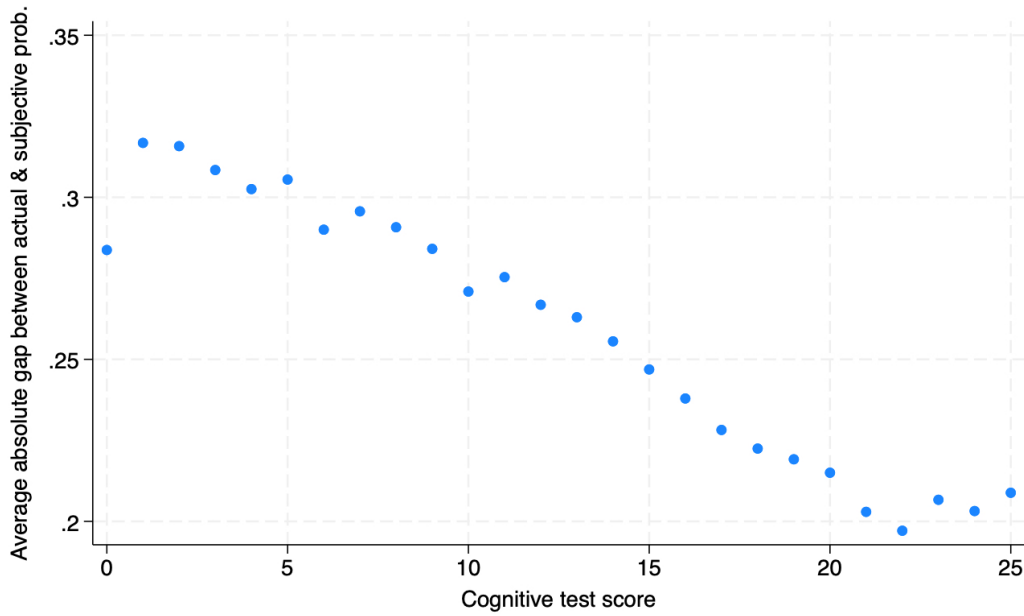


Figure 5 plots the average gap in *actual* vs. *subjective* probabilities against *cognition* scores. As with the Figure 4, due to the large amount of data, only representative averages are presented in Figure 5. The points follow a broadly downward-sloping trend, with no strong indication of nonlinearity (e.g., sharp curvature or threshold effects). Thus, the chosen linear specification appears appropriate. Nevertheless, minor deviations could exist, and future research might incorporate polynomial or spline terms to capture potential nonlinearities.

5.2.2 Homoskedasticity

Homoskedasticity presumes that the error term has a constant variance across all observations. When heteroskedasticity is present, the standard errors from ordinary least squares (or fixed effects) become unreliable, potentially overstating or understating the true precision of estimates.

If certain subgroups (e.g., individuals with particularly high or low *cognition*) exhibit greater variability in the gap than others, the assumption of constant variance is violated. This can distort inference about significance tests and confidence intervals.

The Breusch-Pagan test (Breusch and Pagan (1979)) indicates significant heteroskedasticity ($\chi^2 = 764.11, p < 0.001$). Consequently, using conventional standard errors could lead to incorrect inferences.

To address the heteroskedasticity detected by Breusch-Pagan test, robust standard errors should be employed (White (1980)). These robust standard errors produce valid inference even when error variance is not constant, preserving the consistency of coefficient estimates.

5.2.3 No Serial Correlation in Errors

No serial correlation (autocorrelation) implies that, after controlling for observed variables and fixed effects, the error terms across different time periods for the same individual are uncorrelated. In panel data, violating this assumption can yield downwardly biased standard errors and inflated test statistics.

If an individual's time-varying unobserved factors persist (e.g., a lasting health condition) and simultaneously affect both the error term and *cognition* over consecutive periods, autocorrelation arises and complicates standard inference.

The Wooldridge test (Wooldridge (2002)) reveals statistically significant first-order autocorrelation ($F = 16.824, p < 0.001$), rejecting the null hypothesis of no autocorrelation.

Given both heteroskedasticity and first-order autocorrelation, I adopt *clustered standard errors* at the individual level (Arellano (1987)). Clustering adjusts for within-individual error correlation over time, thus producing consistent standard error estimates under both heteroskedastic and autocorrelated conditions.

5.2.4 No Perfect Multicollinearity

Lastly, the no perfect multicollinearity assumption requires that no explanatory variable can be expressed as an exact linear combination of the others. While some degree of correlation is inevitable in observational data, perfect multicollinearity prevents estimation of unique parameter estimates.

High correlation among regressors can inflate standard errors (multicollinearity). I use the Variance Inflation Factor (VIF) to diagnose multicollinearity (Belsley et al. (1980)). One variable, *iadl*, exhibits a VIF over 10, suggesting strong correlation with *adl* or other functional-status measures.

To mitigate multicollinearity, I exclude *iadl* from the regressions. After its removal, VIF values for remaining variables fall below the conventional thresholds, indicating that problematic collinearity is no longer a concern.

In summary, diagnostic tests indicate that (1) a linear specification is appropriate; (2) heteroskedasticity and serial correlation are present, calling for robust, *clustered standard errors*; and (3) a single variable with a high VIF is dropped to alleviate multicollinearity. Together, these adjustments ensure that the *two-way fixed effects* model provides the most reliable possible estimates for the relationship between *cognition* and the gap in *subjective* versus *actual* health risk perceptions.

5.3 Results

Table 6 presents two regressions of the absolute gap between *subjective* and *actual* health-risk probabilities on the *cognition*. Column (1) excludes fixed effects, while Column (2) incorporates *individual and time (two-way) fixed effects* with *clustered standard errors*.

Table 6: Regression of the Gap of Probabilities on the Cognition

	(1) no fixed effects	(2) fixed effects
Cognition	-0.00566*** (-42.45)	-0.000984*** (-3.57)
Constant	0.330*** (177.89)	0.267*** (71.29)
Observations	114209	88762
R^2	0.0155	0.586

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

In Column (1), the coefficient is -0.00566 (significant at the 1% level), indicating that a one-point increase in the *Cognition* is associated with a 0.57 percentage-point reduction in the gap. When individual-level and time effects are controlled in Column (2), the coefficient decreases in magnitude (to -0.000984), yet remains significant. This decline suggests that part of the initially observed association was explained by unobserved factors constant within individuals or across time.

The R^2 rises from 0.0155 to 0.586 once fixed effects are included, reflecting the considerable explanatory power of time-invariant individual differences and common temporal influences. The large jump in R^2 underscores the importance of controlling for unobserved heterogeneity when investigating the link between *cognition* and misperceived health risks.

Overall, the negative and statistically significant estimates across specifications consistently point to an inverse relationship between *cognition* and the magnitude of risk-assessment errors. Older adults with higher cognitive scores, on average, exhibit a smaller gap between their *subjective* and *actual* health-risk probabilities.

5.4 Robustness Check

Table 7 reports additional specifications designed to test the stability of the main findings from the *two-way fixed effects* model with *clustered standard errors*. Each column adds a different set of control variables, ranging from basic individual attributes (e.g., *Body Mass Index*) to more

comprehensive health and demographic factors (e.g., *Limited in Activities*, *Verbal Fluency Score*). In all cases, the dependent variable is the absolute gap between *subjective* and *actual* health-risk probabilities.

Table 7: Robustness check for Two-way Fixed Effects Model

	(1)	(2)	(3)	(4)	(5)
Cognition	-0.000804*** (-2.61)	-0.00171*** (-3.32)	-0.00172*** (-3.33)	-0.00172*** (-3.33)	-0.00175*** (-3.39)
Body Mass Index	0.0000867 (0.39)	0.000361 (1.06)	0.000363 (1.07)	0.000364 (1.07)	0.000369 (1.09)
Ever Diagnosed	0.00382 (1.43)	0.00220 (0.46)	0.00194 (0.40)	0.00212 (0.44)	0.00177 (0.37)
Taking Medication	-0.00267 (-0.90)	0.00288 (0.57)	0.00435 (0.78)	0.00446 (0.80)	0.00493 (0.88)
Depression Scale	0.00486*** (8.15)	0.00387*** (3.87)	0.00381*** (3.79)	0.00368*** (3.63)	0.00378*** (3.73)
Exercise Frequency	0.000657 (0.80)	0.00152 (1.08)	0.00150 (1.06)	0.00130 (0.92)	0.00117 (0.83)
Limited in Activities	0.00848*** (4.55)	0.0148*** (4.64)	0.0148*** (4.63)	0.0156*** (4.82)	0.0155*** (4.78)
Verbal Fluency Score	-0.000632*** (-3.71)	-0.000604** (-2.14)	-0.000608** (-2.16)	-0.000597** (-2.12)	-0.000601** (-2.14)
Constant	0.242*** (25.23)	0.248*** (10.15)	0.245*** (9.93)	0.243*** (9.77)	0.241*** (8.98)
Controls	7	30	43	54	68
Observations	86854	37604	37604	37604	37604
R^2	0.588	0.650	0.650	0.651	0.651

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Across Columns (1) – (5), the coefficient on *Cognition* remains negative and statistically signifi-

cant at the 1% level. These estimates range from approximately -0.0008 to -0.0017 , indicating that higher cognitive function consistently predicts a smaller gap in perceived health risks, even after controlling for a wide array of covariates. This consistency reinforces the key finding that *cognition* plays a pivotal role in aligning *subjective* probabilities with *actual* measures.

Although *Body Mass Index* (BMI) is commonly associated with health outcomes, its coefficients here are small and generally not statistically significant. This result suggests that, net of other variables (e.g., chronic conditions or physical limitations), BMI alone may not markedly affect how older adults perceive their health risks.

The variable *Ever Diagnosed* indicates whether the respondent has ever received a diagnosis of a major health condition. The coefficients tend to be small and often lack statistical significance. One plausible interpretation is that having a prior diagnosis—at least among older adults—does not necessarily translate into systematically higher or lower misperceptions of risk, once depression, medication use, and other functional measures are taken into account.

Similar to *Ever Diagnosed*, the effect of current medication use remains relatively weak in most columns. Older individuals on medication may already be managing known conditions and might therefore have more informed risk perceptions—or, conversely, might underestimate new risks. However, the lack of statistical significance implies that medication status, by itself, does not strongly predict errors in risk assessment.

In nearly all specifications, depression emerges as positively and significantly correlated with the gap, suggesting that individuals with more depressive symptoms tend to overestimate or underestimate their health risks to a larger extent. Psychologically, higher depression may distort one's self-perception, leading to either pessimistic or inconsistent evaluations of one's mortality risk.

Although regular exercise is typically associated with better physical and mental health, its coefficient here is small and often not statistically significant. This finding suggests that, while exercise likely benefits overall health, it may not substantially reduce the mismatch between *subjective* and *actual* health-risk probabilities—at least not in isolation from broader lifestyle and psychological factors.

The variable *Limited in Activities*, often related to the difficulty or inability to perform daily tasks, consistently shows a positive association with the gap. Older adults who are limited in daily activities appear more prone to misjudging their health risks. One possibility is that experiencing functional limitations prompts either an exaggerated sense of vulnerability or an underestimation of future deterioration, both of which widen the gap between *subjective* and *actual* risk estimates.

Verbal fluency Score is indicative of certain cognitive and language-processing abilities. The negative and significant coefficients on this variable imply that older adults who score higher in verbal fluency tend to assess their health risks more accurately. This aligns well with the broader conclusion that stronger cognitive skills—in both general and language-specific domains—are linked

to lower misalignment in perceived risk.

In all models, the constant term remains positive and significant, reflecting the baseline gap in risk perception among individuals with average or reference-level covariates. The varying number of control variables across columns changes the sample size and marginally affects model fit (R^2), but in each specification the main relationship between *Cognition* and the *Probabilities Gap* remains robust.

Overall, these expanded specifications verify that the negative relationship between *cognition* and the health-risk gap holds under numerous model configurations. While a few control variables—particularly *Depression Scale*, *Limited in Activities*, and *Verbal Fluency Score*—exhibit consistent links to misperceived risks, others show weaker or no significant effects. The persistence of the main coefficient for *cognition* underscores the potential value of interventions aimed at enhancing cognitive abilities in older adults, not only for day-to-day functioning but also for more accurate health-related decision-making. Such findings may inform policies and programs designed to reduce risk misperception and improve overall well-being in aging populations.

5.5 Heterogeneity Analysis

5.5.1 Age Heterogeneity

Table 8 examines whether the effect of *Cognition* on miscalibrated health-risk assessments varies across different age groups. Columns (1), (2), (3), and (4) correspond to individuals aged 50–59, 60–69, 70–79, and 80+, respectively.

Table 8: Heterogeneity Analysis - Age

	(1) age 50	(2) age 60	(3) age 70	(4) age 80
Cognition	-0.00126 (-1.17)	-0.000595 (-0.67)	-0.00337*** (-2.83)	-0.00430* (-1.81)
Body Mass Index	0.000524 (0.92)	-0.000327 (-0.53)	0.00116 (1.21)	-0.000346 (-0.29)
Ever Diagnosed	0.000596 (0.06)	0.000717 (0.08)	0.0158 (1.17)	0.0137 (0.47)
Taking Medication	0.00698 (0.64)	-0.0123 (-1.23)	0.0138 (0.92)	-0.00797 (-0.26)
Depression Scale	0.00425** (2.04)	0.00309* (1.69)	0.00155 (0.65)	0.00221 (0.58)
Exercise Frequency	-0.00439 (-1.41)	-0.00305 (-1.22)	0.00719** (2.19)	-0.000329 (-0.05)
Limited in Activities	0.0240*** (3.24)	-0.000951 (-0.17)	0.0155** (2.12)	0.0482*** (3.88)
Verbal Fluency Score	-0.000927* (-1.68)	-0.000367 (-0.80)	-0.000935 (-1.19)	0.00139 (0.92)
Constant	0.168*** (2.83)	0.238*** (4.87)	0.349*** (5.81)	0.324*** (3.43)
Controls	68	68	68	68
Observations	7906	10643	7449	2724
R^2	0.673	0.645	0.636	0.633

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

In Columns (1) and (2), the coefficient on *Cognition* is negative but statistically insignificant, suggesting that among younger participants, variations in cognitive performance do not have a robust association with the gap between *subjective* and *actual* risks. By contrast, in Columns (3) and (4), for individuals age 70+, the coefficients are both larger in magnitude and statistically significant. These

findings imply that cognitive ability becomes more consequential for risk perception at advanced ages, possibly because declining fluid intelligence and greater health uncertainty accentuate the role of *cognition* in accurately gauging mortality risks.

Depression Scale exhibits a positive and (in some cases) significant association with the gap at younger ages (e.g., Column (1)), but this effect diminishes in older cohorts. One explanation could be that, for those in their early fifties, depression strongly distorts self-assessed health risks; however, for seniors in their seventies or eighties, other factors (e.g., comorbidities, functional limitations) may overshadow the role of depressive symptoms in shaping subjective perceptions.

Interestingly, *Limited in Activities* show a significant positive coefficient only at certain ages—strong in Column (1) (age 50–59) and Column (4) (age 80+). This pattern suggests a possible U-shaped relationship: at relatively younger ages, functional limitations may serve as an unusual or alarming sign, fueling overestimation of risk, whereas in very old age, extensive health limitations might lead to either pessimistic or inconsistent judgments about longevity.

The coefficient on *Verbal Fluency Score* is negative in several columns, although its significance varies across age groups. This reinforces the broader finding that stronger cognitive and language skills tend to correlate with more accurate risk assessments, but the effect may not be uniform over the life course.

The results highlight that the influence of *cognition* on health-risk miscalibration intensifies at older ages. Younger adults (age 50–69) may rely less on formal *cognition* scores when judging health outcomes or might benefit from better baseline health, masking cognitive effects. By age 70+, cognitive declines appear to have a more pronounced impact on one’s ability to align *subjective* beliefs with *actual* risks.

5.5.2 Probability Heterogeneity

Table 9 explores whether *cognition*’s effect depends on the magnitude of the *actual* probability of mortality. The table divides the sample into groups with a predicted risk of 0–20%, 21–40%, 41–60%, and 61–100%.

Table 9: Heterogeneity Analysis - Actual Probability

	(1)	(2)	(3)	(4)
	prob. 20%	prob. 40%	prob. 60%	prob. 100%
Cognition	-0.00185*** (-2.84)	-0.00138 (-0.62)	-0.00714 (-1.47)	-0.00360 (-1.23)
Body Mass Index	0.00123*** (2.64)	-0.00165 (-1.13)	-0.000328 (-0.14)	-0.000129 (-0.09)
Ever Diagnosed	0.00380 (0.66)	0.0412 (0.85)	0 (.)	-0.00707 (-0.06)
Taking Medication	0.0156** (2.42)	-0.00544 (-0.17)	-0.274* (-1.85)	0.0435 (0.46)
Depression Scale	0.00514*** (3.79)	0.00924** (2.53)	-0.000331 (-0.05)	-0.00714 (-1.52)
Exercise Frequency	0.000914 (0.52)	0.00667 (1.17)	-0.000383 (-0.03)	-0.00972 (-0.84)
Limited in Activities	0.0143*** (3.07)	0.00453 (0.37)	0.00524 (0.24)	-0.0226 (-1.02)
Verbal Fluency Score	-0.000346 (-0.96)	-0.00161 (-1.49)	-0.00134 (-0.41)	0.000569 (0.22)
Constant	0.180*** (4.56)	0.455*** (4.17)	0.195 (1.08)	0.211* (1.74)
Controls	68	68	68	68
Observations	21470	1762	452	896
R^2	0.703	0.684	0.709	0.697

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

In Column (1) (low *actual* risk of less than 20%), the coefficient on *Cognition* is negative and highly significant, indicating that better cognitive function substantially reduces the gap at relatively low risk levels. However, in Columns (2) through (4), representing higher predicted probabilities (over 20%), the coefficient is smaller in magnitude and not consistently significant. One possibility

is that when the likelihood of mortality becomes moderate to high, older adults' perceptions may already be heavily influenced by known chronic conditions or functional limitations, lessening the marginal role of *cognition* in shaping risk estimates.

In the low-probability group (Column (1)), *Body Mass Index* (BMI) is positively and significantly associated with the gap, suggesting that individuals in relatively good health but with higher BMI may overestimate their mortality risk. At higher probabilities, the effect diminishes, possibly because at that stage, more pressing health factors overshadow weight-based cues in forming risk perceptions.

Taking Medication shows inconsistent effects, sometimes positive (e.g., Column (1)) and sometimes negative or insignificant (e.g., Column (3)). This variability may reflect differing types of medications or underlying health conditions. In lower-risk scenarios, *Taking Medication* might heighten sensitivity to potential risks, while in advanced disease states, medication use may neither reassure nor alarm individuals sufficiently to change their risk perceptions.

Depression Scale tends to increase the gap in Columns (1) and (2), indicating that at low and moderate probabilities, depressive symptoms could distort risk perceptions upward or downward. But at the highest risk level (Column (4), probability 61–100%), its effect vanishes, consistent with the idea that once mortality risk is almost certain, psychological states play a smaller role in how individuals gauge their situation.

Being limited in activities appears to exacerbate risk misjudgments primarily at lower predicted probabilities (e.g., Column (1)), while verbal fluency does not show a strong effect in any particular probability group. This suggests that functional impairments might trigger disproportionate concern when the *actual* probability is still modest, whereas high-probability contexts leave less room for subjective miscalibration based on functional limitations or language abilities.

The results indicate that *cognition*'s mitigating impact on miscalibration is especially pronounced when the *actual* probability of mortality is relatively low. As *actual* risk escalates, other health factors—such as advanced chronic conditions or activity limitations—may overshadow *cognition*'s role in shaping beliefs. These patterns underscore the importance of tailoring interventions or information campaigns according to individuals' *actual* risk levels. In particular, helping older adults with lower *actual* risks interpret their health status more accurately may yield the greatest improvements in risk alignment, while in high-risk contexts, targeted interventions addressing specific conditions or functional deficits might be necessary.

6 Conclusion

This study examined the impact of cognitive decline on older adults' biases in their assessment of their own health risks. Specifically, the study demonstrated a strong negative correlation between cognitive ability and older adults' bias in their assessment of their own health risks, i.e., as cognitive

ability declines, relatively healthy older adults overestimate their health risks more, whereas older adults in poorer health underestimate their health risks more.

This study is divided into three main steps. First, in order to obtain the bias values of older adults' assessment of their health risks, I need to obtain two data first: *actual* health risk and *subjective* health risk. *Subjective* health risk can be found in the dataset I used, Survey of Health, Ageing and Retirement in Europe (SHARE), but there is no direct *actual* health risk in this dataset, so further estimation is needed. Specifically, I estimated the *actual* health risk probability using several hundred health-related variables and death status after 4 years, and the specific estimation model was BioBERT, a transformer-based deep learning model that currently has the best prediction performance, and obtained better performing estimates (e.g., the overall Accuracy was 86.83%, and the Precision was 96% for the elderly who were still alive after 4 years). precision of 96.69% for older adults).

Second, after obtaining data on *actual* and *subjective* health risks, I plotted the relationship between the two and provided empirical validation of the theory of the *probability weighting function*. Specifically, healthier older adults overestimate their health risks, older adults in fair health estimate their health risks closer to their *actual* risks, and older adults in poorer health significantly underestimate their health risks, which is consistent with the *probability weighting function*.

Third, to obtain data on cognitive ability pairs of older adults, I assessed cognitive ability using SHARE data based on the way cognitive functioning is measured by the Health and Retirement Study (HRS), one of the best studies done in the field of aging. Subsequently, I did a regression analysis of the effect of the bias of older adults' assessment of their own health risks on the cognitive functioning using a *two-way fixed effects* model with *clustered standard errors* and came up with significant results (significance level of 0.001). By adding different number of control variables at different specifications, I verified the robustness of the regression results. In addition, I grouped the data according to age and did the analysis of heterogeneity, which showed that the effect of cognitive ability on health risk assessment bias was more significant in older older adults (70 and above), while this effect was not significant in younger older adults (50–69 years) this effect was not significant. This may be due to the fact that there is more cognitive decline in relatively older adults, and only when cognitive decline reaches a certain level does it have a significant effect on health risk assessment bias. I also did a heterogeneity analysis for *actual* health risk, and only older adults with lower health risk (less than 20%) had a more significant effect of cognitive ability on health risk assessment bias. One possibility is that by the time the likelihood of death is moderate to high, older adults' *cognition* may already be severely affected by known chronic conditions or functional limitations, reducing the marginal role of *cognition* in shaping risk estimates.

This study contributes to our understanding of the role cognitive ability plays in the process of decision making. Cognitive decline is a natural process and the rate of cognitive decline is faster in

old age. If cognitive decline leads to greater bias in older adults' assessment of their own health risks, then this will directly affect older adults' health decisions. For example, older adults with better health and more cognitive decline may be more focused on the importance of healthy lifestyle habits; they may be more likely to choose a nutritionally balanced diet, more likely to increase the frequency of exercise, be more active in going to the hospital for treatments, and consume more of the medications they need, and so on. In addition, when it comes to the choice of medical insurance, they may opt for more expensive insurance packages with higher disease coverage.

7 Discussion

Building on the findings above, several practical implications arise for healthcare policy, insurance markets, and geriatric care. First, cognitive decline can lead to an overestimation of health risks among older adults in better health, which in turn may lead to inefficient use of resources. While preventive health behaviors—such as balanced diets, increased physical exercise, and frequent medical checkups—are generally beneficial, an excessive focus on perceived risks might impose unnecessary financial burdens. Over-insuring, for example, could divert resources away from other essential expenses, especially for those on fixed incomes.

Second, for older adults in poorer health, cognitive decline can lead to underestimation of health risks, which in turn may lead to more serious consequences. Delays in seeking medical attention, insufficient insurance coverage, or a lack of preventive measures can exacerbate existing conditions and lead to deteriorating health over time. Awareness campaigns or interventions tailored to older adults with declining cognitive ability could be designed to correct these underestimations. Programs might include simplified risk assessment tools, targeted health education, or community-based support systems that encourage timely and adequate health service utilization.

These observations align with a broader literature on the multifaceted impact of cognitive decline in older adults. Extensive research in gerontology and behavioral economics has demonstrated that declining cognitive function can impair decision-making in various domains. For example, some studies show that older adults may become more risk-averse or struggle to incorporate new information into financial decisions, leaving them vulnerable to poor investment choices or even financial exploitation (Boyle et al. (2013); Agarwal et al. (2009)). In the realm of health behaviors, cognitive decline has been linked to lower medication adherence and inadequate self-care routines, as older adults may fail to remember dosing schedules or the importance of preventive measures (Murray et al. (2004)). Moreover, scholars have identified that age-related cognitive changes can affect emotional regulation and risk perception, potentially leading to suboptimal trade-offs between immediate costs and long-term benefits in healthcare decisions (Scheibe and Carstensen (2010)). By isolating how cognitive ability specifically distorts health-risk assessments, this research

situates itself within a growing discourse on how best to support older populations through tailored interventions and informed policy measures.

In terms of policy and healthcare design, this thesis underscores the importance of incorporating cognitive screening measures in regular medical checkups for the elderly. Healthcare practitioners, insurers, and policymakers might collaborate to develop personalized interventions. For instance, simplified insurance information—presented in clear, accessible language—could help those with reduced cognitive ability make more informed decisions, thereby mitigating the risk of under- or over-insurance.

Overall, this study integrates into a substantial literature focused on mitigating the negative consequences of cognitive decline. By highlighting the mechanisms behind health-risk misperceptions, it offers evidence-based insights that can inform the design of more effective interventions and policies. Future research could expand these findings by exploring how social support networks, psychological factors (e.g., anxiety and depression), and digital health technologies might further shape the relationship between cognitive ability and health-risk assessment.

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